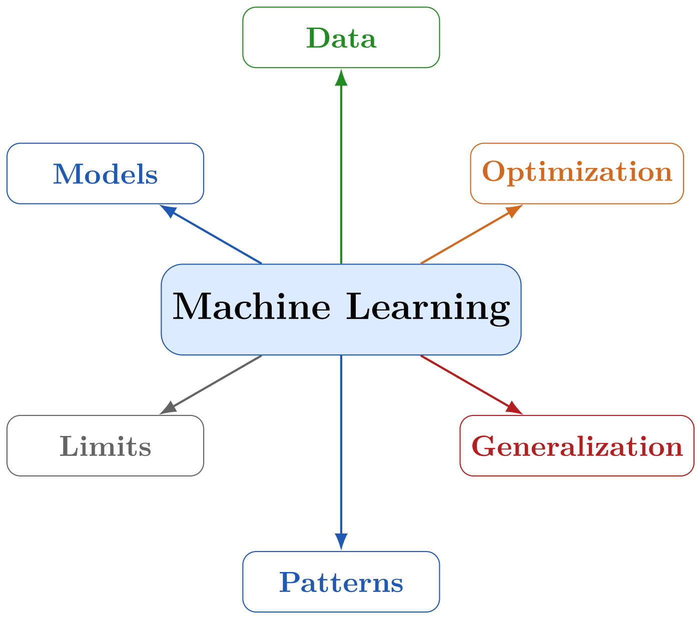
Made by Farhan Shahriyar

Machine Learning:
Concepts & Critical Questions

A Comprehensive Analytical Study

Covering Learning, Optimization, Understanding, Generalization, Overfitting, Underfitting, and Human–Machine Cognition Comparison



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Q1. Definition of "Learning" in ML vs. Human Learning

Question: Define "learning" in the context of machine learning. How does it differ from human learning?

Machine Learning Definition

In machine learning, **learning** is the iterative adjustment of internal parameters (weights/coefficients) to minimise a predefined loss function over a training dataset. Given $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ and model f_θ, learning solves:

$$\theta^* = \arg\min_\theta \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}(f_\theta(x_i),\, y_i)$$

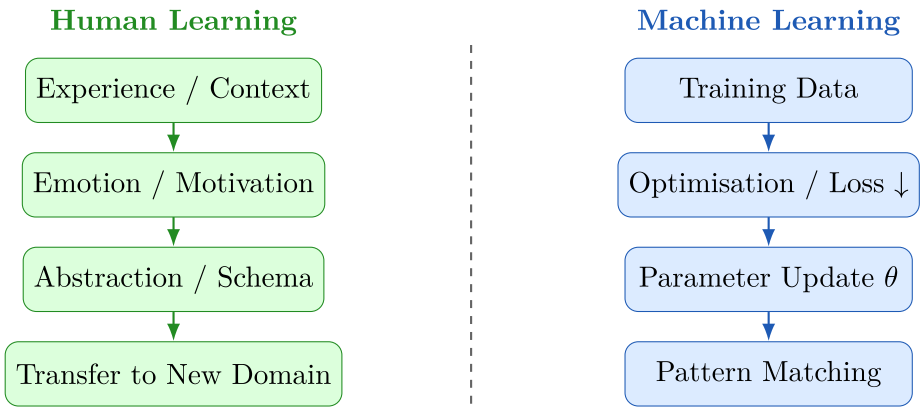
No consciousness, intent, or understanding is involved—only numerical adjustment.

Human Learning Definition

Human learning is a **cognitive and biological process** involving acquisition, encoding, storage, and retrieval of knowledge through experience, reasoning, emotion, and social interaction. It draws on existing schemas, can operate from a single example, and forms conceptual abstractions tied to real-world meaning.

Key Differences

Dimension	Machine Learning	Human Learning
Mechanism	Gradient-based optimisation	Synaptic plasticity, emotion, cognition
Data need	Large labelled datasets	Few examples suffice
Adaptability	Bounded by training dist.	Flexible, cross-domain
Understanding	Statistical correlation only	Causal, conceptual
Motivation	None (external objective)	Curiosity, goals
Transfer	Poor across tasks	Natural across domains
Error correction	Requires retraining	Self-reflective, immediate


Human Learning
Experience / Context
Emotion / Motivation
Abstraction / Schema
Transfer to New Domain
Machine Learning
Training Data
Optimisation / Loss ↓
Parameter Update θ
Pattern Matching

Q2. Why "Learning" in ML is Metaphorical

Question: Explain why the term "learning" in "machine learning" is considered metaphorical.

The word **learning** in everyday language implies:

(i) *Comprehension*—grasping the meaning of new information;

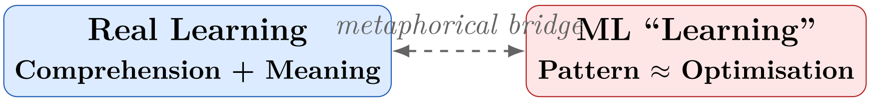
(ii) *Intentionality*—a conscious decision to acquire knowledge;

(iii) *Retention and recall*—flexible storage and retrieval;

(iv) *Causal reasoning*—understanding *why* something is true.

Machine learning satisfies *none* of these in the strict philosophical sense. Arthur Samuel (1959) coined "machine learning" as a pragmatic shorthand for "programs that improve performance with experience." Philosophers and AI researchers (notably Searle, Dreyfus) have argued there is no genuine cognitive overlap.

ML "Learning" = Iterative numerical optimisation of parameters such that a mathematical error metric decreases on training samples.



Similar outputs, fundamentally different internal processes

Q3. Role of Optimisation in Machine Learning

Question: What is the role of optimisation in the learning process of a machine learning model?

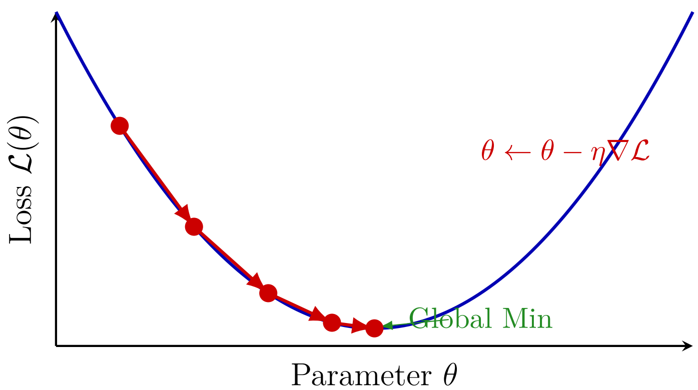
Optimisation is the engine of machine learning. The dominant algorithm is **gradient descent**:

$$\theta_{t+1} = \theta_t - \eta \, \nabla_\theta \, \mathcal{L}(\theta_t)$$

where η is the learning rate and $\nabla_\theta \mathcal{L}$ is the gradient of the loss with respect to the parameters.

- **Defines the objective:** The loss encodes what to improve (e.g. cross-entropy for classification).
- **Drives parameter updates:** Each step reduces prediction error.
- **Determines generalisation:** Regularisation (L1/L2, dropout) prevents memorisation.
- **Controls speed and stability:** Optimizer choice (SGD, Adam, RMSProp) governs convergence.

Gradient Descent on a Loss Surface

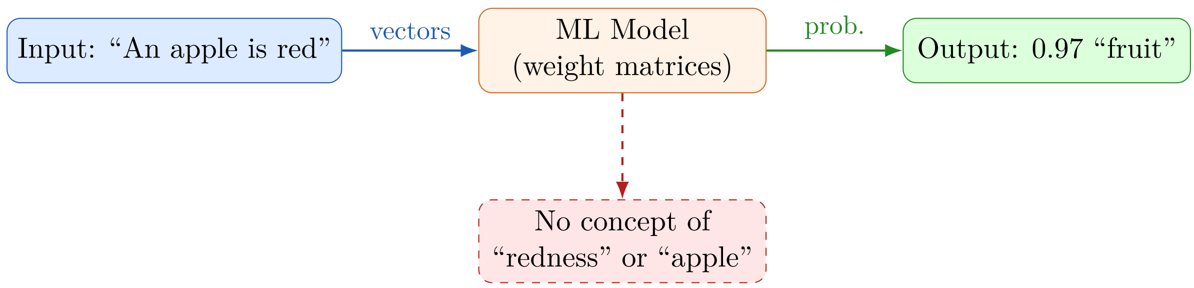


Q4. Why ML Models Cannot "Understand" Data

Question: Why can machine learning models not be said to "understand" the data they are trained on?

Understanding requires semantic grounding—connecting symbols to real-world meaning and causal structure. ML models lack this entirely:

1. **Syntax, not Semantics.** A model processes patterns of numbers, not meaning. "Cat" is a floating-point vector, not a concept linked to a living animal.
2. **No Causal Model.** Models learn conditional distributions $P(y \mid x)$. They cannot answer "Why does x cause y?"
3. **Searle's Chinese Room (1980).** A system can produce correct outputs by following rules without understanding what the symbols mean—precisely what ML models do.
4. **Brittleness.** Tiny adversarial perturbations cause catastrophic failure, revealing superficial pattern matching.
5. **No Embodied Experience.** Human concepts are grounded in sensorimotor experience; ML models have only numerical co-occurrence statistics.



Q5. Impact of Data Quantity and Quality on Learning

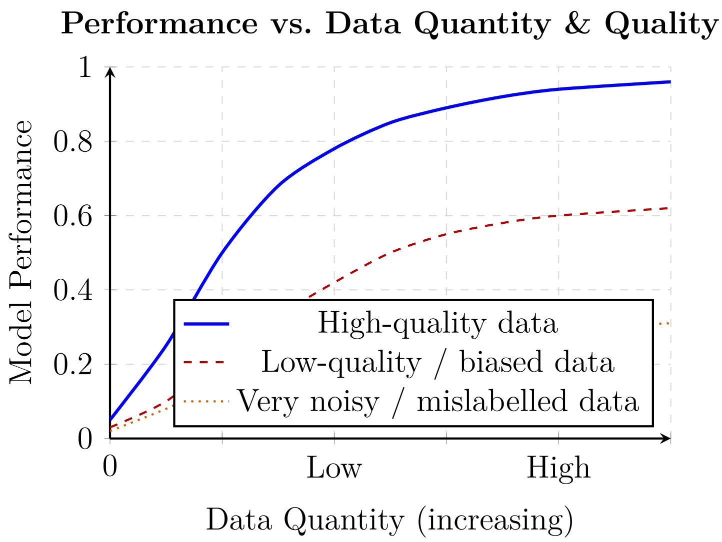
Question: How does data quantity and quality affect the kind of "learning" achieved by a machine?

Data Quantity

- **Too little** ⇒ *underfitting*: the model cannot capture the underlying distribution.
- **Sufficient and diverse** ⇒ generalizable patterns can be extracted.
- **Large but non-diverse** ⇒ biases are amplified rather than corrected.

Data Quality

- **Noisy labels** ⇒ incorrect mappings are learned.
- **Biased sampling** ⇒ model reflects societal/measurement biases (garbage in, garbage out).
- **Class imbalance** ⇒ minority classes are ignored.
- **Irrelevant features** ⇒ spurious correlations learned instead of causal relationships.



Q6. Fruit Classifier Case Study: Mango Failure

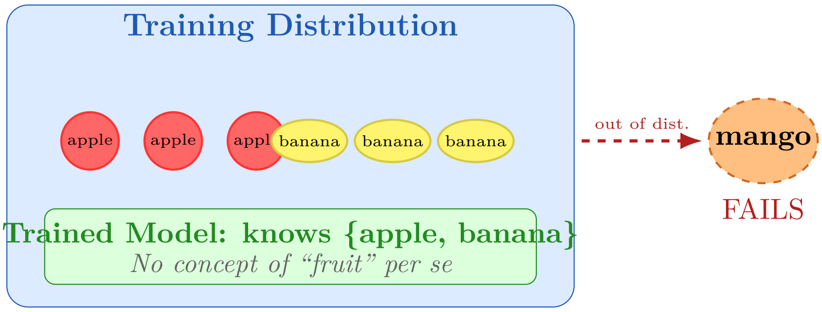
Question: You train an image classifier on fruits. It performs well on apples and bananas but fails on mangoes. What does this reveal about the nature of "learning"?

1. **Learning is distribution memorisation.** The model learned to associate specific pixel patterns with labels, not "what a fruit is."

2. **No semantic generalisation.** A child recognises a mango as a fruit via conceptual reasoning; the model has no such concept.

3. **Closed-world assumption.** ML classifiers assume test distribution ≈ training distribution. Mangoes violate this.

4. **Remedy requires data, not reasoning.** The fix is to add mango examples and retrain—not to teach a concept.



Q7. Does a Chatbot Understand Language?

Question: A chatbot is trained on millions of text conversations. Does it understand language? Justify using the principles of machine learning.

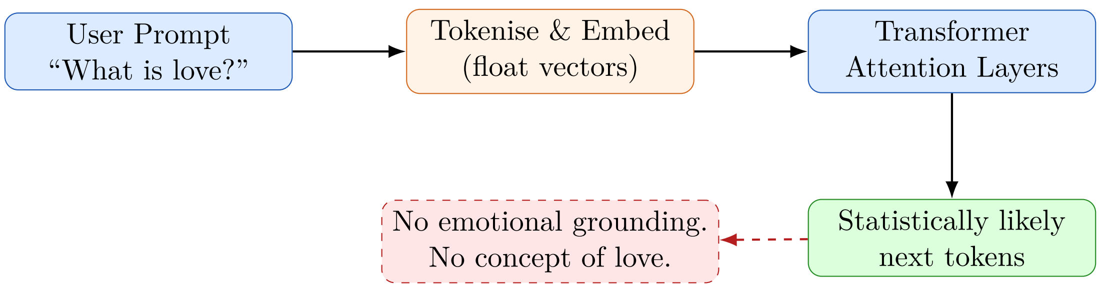
No. A chatbot performs sophisticated **statistical next-token prediction**:

$$P(x_{n+1} \mid x_1, \dots, x_n) = \text{softmax}(W \cdot h_n)$$

where h_n is the hidden state. It selects the statistically most likely continuation—not the *meaningful* one.

Evidence Against Understanding

- **Hallucinations:** Confident false statements arise because token plausibility $\neq$ factual truth.
- **Inconsistency:** The same question phrased differently yields contradictory answers.
- **No grounding:** "Pain" is a token embedding; the model has never experienced pain.
- **Adversarial fragility:** Simple rephrasing completely changes responses—a sign of pattern matching, not comprehension.
- **Searle's Chinese Room:** Syntactic rule-following is not semantic comprehension.



Q8. Rebuttal: "Machines Learn Just Like Humans"

Question: A student argues: "Since a machine can improve from experience, it learns just like a human." How would you respond using technical reasoning?

The student conflates **behavioural similarity** with **mechanistic equivalence**. A thermometer adjusts its reading with temperature, yet no one says it learns.

1. **Different definitions of "experience":** ML experience = labelled training data. Human experience = unstructured, sensory, emotional, social signals.
2. **Different mechanisms:** ML = gradient descent on a mathematical objective. Human = synaptic plasticity, metacognition, executive function.
3. **Catastrophic forgetting:** Learning a new ML task overwrites parameters for old ones. Humans integrate new and old knowledge naturally.
4. **Sample efficiency gap:** A child recognises a dog from 2–3 examples; an ImageNet classifier needs millions of images.
5. **No metacognition:** ML models have no self-awareness of their own knowledge gaps without external uncertainty quantification.
6. **No intrinsic motivation:** The machine does not *want* to improve; the objective is externally imposed.

Conclusion: Improvement from experience is necessary but *not sufficient* for equivalence with human learning. The student's claim is a surface-level analogy.

Q9. RL Chess Agent: Does It Understand Strategy?

Question: A reinforcement learning agent learns to play chess better than humans. Has it learned to understand chess strategy? Discuss.

What RL Does

The agent learns a policy $\pi_\theta(a \mid s)$ by maximising cumulative reward:

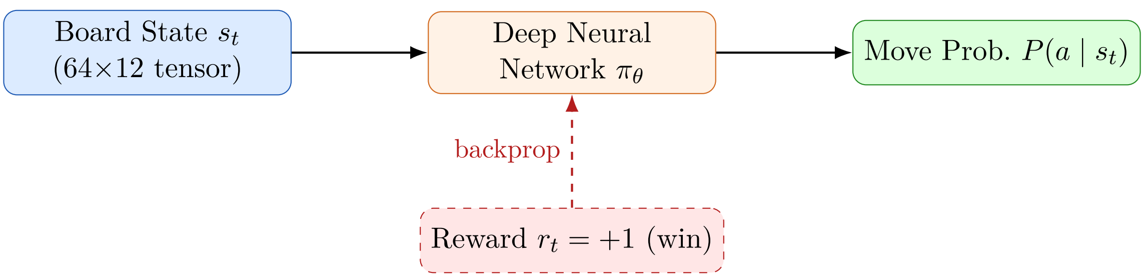
$$\theta^* = \arg\max_\theta \ \mathbb{E}_{\tau \sim \pi_\theta}\left[\sum_{t=0}^{T} \gamma^t r_t\right]$$

AlphaZero (DeepMind) achieved superhuman performance via pure self-play—no human-coded rules.

Performance ≠ Strategic Understanding

- **What it does:** Maps board-state tensors to move probabilities via a deep value network.
- **What it cannot do:** Explain *why* a move is strategically superior. Grandmasters articulate plans; AlphaZero cannot.
- **No context transfer:** Remove one rule and the agent must retrain from scratch. A human adapts immediately.
- **No intentionality:** The agent does not "want" to win; it maximises a reward signal.

Key insight: Superhuman performance is an emergent property of scale and optimisation—not comprehension. The agent holds an *approximation of optimal play* in its weights, not an abstract understanding of strategy.



No "strategy" concept stored anywhere

Q10. Overfitting and Underfitting: Limits of Machine Learning

Question: Explain how overfitting and underfitting illustrate the limits of what machines can "learn."

Overfitting and underfitting are two sides of the **bias–variance tradeoff**, jointly revealing that machine "learning" is a delicate statistical balancing act.

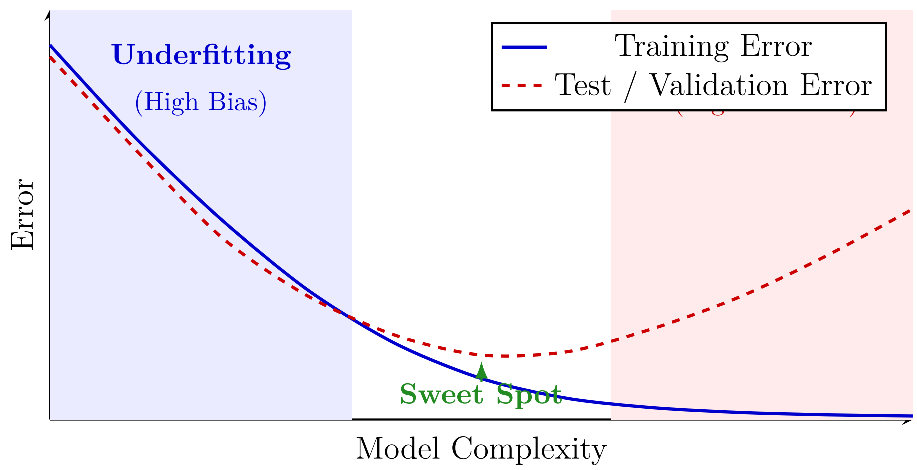
Underfitting (High Bias)

- High training *and* test error.
- Model too simple to extract basic regularities.
- Illustrates: *Insufficient capacity ⇒ learning is incomplete.*

Overfitting (High Variance)

- Very low training error, high test error.
- Model memorised noise and irrelevant patterns.
- Illustrates: *"Learning" can be brittle memorisation, not insight.*

Bias–Variance Tradeoff: Overfitting vs. Underfitting



Philosophical limit: A human who "over-studies" can still generalise using judgment. A machine that overfits is completely trapped in its training distribution—this asymmetry is fundamental.

Q11. Human Cognition vs. Machine Learning

Question: Compare and contrast human cognitive learning and machine learning from the perspectives of generalisation, abstraction, and adaptability.

Generalisation

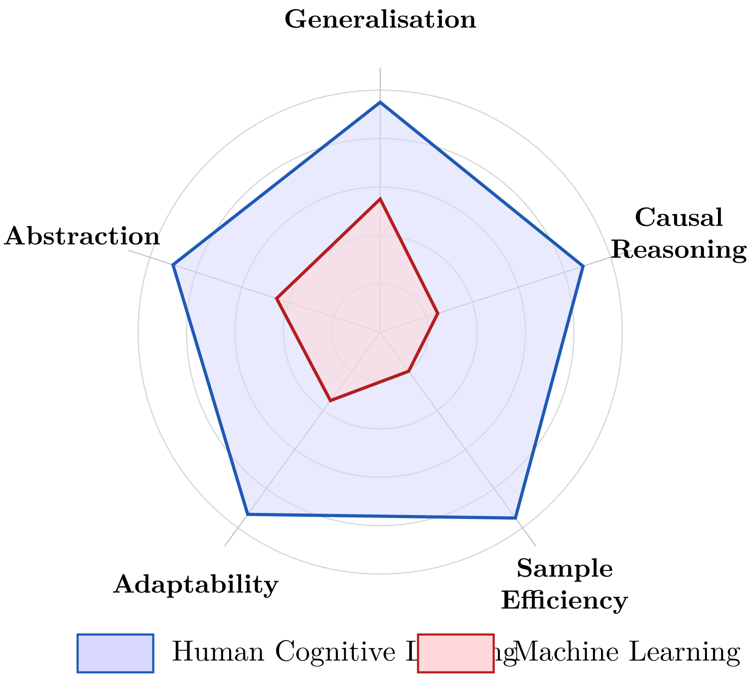
- **Humans:** From few examples via analogical reasoning and prior knowledge.
- **Machines:** Require large datasets; degrade sharply outside training distribution.

Abstraction

- **Humans:** Form grounded concepts ("justice", "infinity") operable at multiple levels simultaneously.
- **Machines:** Distributed embeddings encode statistical regularities—not true, introspectable abstractions.

Adaptability

- **Humans:** Zero-shot and cross-domain transfer is natural; improvisation is effortless.
- **Machines:** Suffer catastrophic forgetting; continual learning remains an open research problem.



Dimension	Human Cognition	Machine Learning
Generalisation	Few examples, cross-domain	Large datasets, distribution-bounded
Abstraction	Conceptual, grounded, multi-level	Statistical embeddings; not introspectable
Adaptability	Rapid, zero-shot, creative	Slow; catastrophic forgetting; retraining needed
Sample Eff.	1–10 examples often enough	Thousands to millions required
Causal Reas.	Natural; "why" understood	Correlation only; no causation

Overall Takeaway: Machine learning is a powerful statistical tool that *mimics* aspects of intelligent behaviour through optimisation. It is not equivalent to human learning in mechanism, depth, or flexibility. The terminology is a productive metaphor, but treating it as literal leads to fundamental misunderstandings about what AI systems can and cannot do.